

2026.9.17.

Last time

THM (First isom theorem). If  $\phi: G \rightarrow H$  group homomorphism,  
then  $\ker \phi \triangleleft G$  and  $G/\ker \phi \cong \phi(G)$ .

THM (Second isom theorem). Let  $G$  be a group and let  $H, K$  be subgroups of  $G$   
with  $K$  normal ( $H < G, K \triangleleft G$ ), then  $HK$  is a subgroup of  $G$ ,  
*already shown*

$$K \triangleleft HK, H \cap K \triangleleft H \text{ and } (HK)/K \cong H/(H \cap K)$$

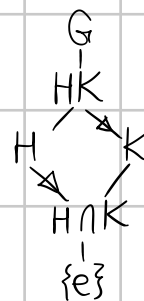
(analogous to  $W_1, W_2$  linear subspaces of  $V$ , then  $(W_1 + W_2)/W_2 \cong W_1/W_1 \cap W_2$ )

Pf. First show  $K \triangleleft HK$  (in fact obvious since  $K \triangleleft G$ ).

Then define a homomorphism  $\phi: H \rightarrow (HK)/K$   
$$\begin{array}{ccc} H & \xrightarrow{\phi} & (HK)/K \\ h & \mapsto & hK \end{array}$$

$\phi$  surjective since  $hK = hK = \phi(h)$

$$\ker \phi = \{h \in H \mid hK = K\} = H \cap K \text{ in particular } H \cap K \triangleleft H.$$
$$\begin{array}{c} \updownarrow \\ h \in K \end{array}$$



By the first isom theorem,  $H/(H \cap K) \cong (HK)/K$ .

REM 1). Clear from the Pf. that instead of  $K \triangleleft G$ , we only need that

$H$  normalizes  $K$  (ie.  $\forall h \in H, hKh^{-1} = K$  ( $hK = Kh$ )).

2). Common way to prove  $G/H \cong G'/H'$

- construct a homomorphism  $G \rightarrow G'$
- compose to get  $G \rightarrow G' \rightarrow G'/H'$
- show surjectivity & compute kernel.

THM (Third isom theorem). Let  $G$  be a group and let  $H, K$  be normal groups of  $G$   
with  $K < H$  ( $\Rightarrow K \triangleleft H$ ), then  $H/K \triangleleft G/K$  and  $(G/K)/(H/K) \cong G/H$ .

(if we write  $\bar{G}$  for  $G/K$ , this reads  $\bar{G}/\bar{H} \cong G/H$ ).

Pf. Consider  $\phi: G/K \rightarrow G/H$ .

$$gK \mapsto gH$$

$\phi$  is well-defined: if  $g_1K = g_2K$  then  $g_1^{-1}g_2 \in K \leq H \Rightarrow g_1H = g_2H$ .

$\phi$  is a homomorphism:  $\phi(g_1K \cdot g_2K) = \phi(g_1g_2K) = g_1g_2H = g_1H \cdot g_2H = \phi(g_1K) \cdot \phi(g_2K)$ .

$\phi$  is surjective: obvious.

$$\ker \phi = \{gK \mid gH = H\} = H/K \Rightarrow \text{in particular } H/K \triangleleft G/K$$

$$\updownarrow$$

$$g \in H$$

By the first isom theorem  $(G/K)/(H/K) \cong G/H$ .

THM (Fourth and lattice isom theorem). Let  $G$  be a group and  $N \triangleleft G$ ,

then there is a natural bijection ( $\pi: G \rightarrow G/N$  is the natural projection)

$$\begin{array}{ccc} \{\text{subgroups of } G \text{ containing } N\} & \begin{array}{c} H \\ \downarrow \\ H/N \end{array} & \begin{array}{c} \pi(H) \\ \uparrow \\ \bar{H} \end{array} \\ \updownarrow & & \\ \{\text{subgroups of } G/N\} & & \end{array}$$

which preserves: inclusion, index of subgroups, intersections, normality, ...

Pf. exercise.

$$\left( \begin{array}{c} \text{lattices of subgroups of } G \text{ containing } N \\ \updownarrow \\ \text{lattices of subgroups of } G/N \end{array} \right)$$

REM useful viewpoint on homomorphisms from a quotient group.

Let  $\phi: G \rightarrow H$  be a group homomorphism and let  $N \triangleleft G$ .

Want: define  $\bar{\phi}: G/N \rightarrow H$  by

$$gN \mapsto \phi(g)$$

Such  $\bar{\phi}$  is well-defined  $\Leftrightarrow N \subset \ker \phi$ .

When  $N \subset \ker \phi$  we say that  $\phi: G \rightarrow H$  factors through  $G/N$ .

商群的泛性质 universal property

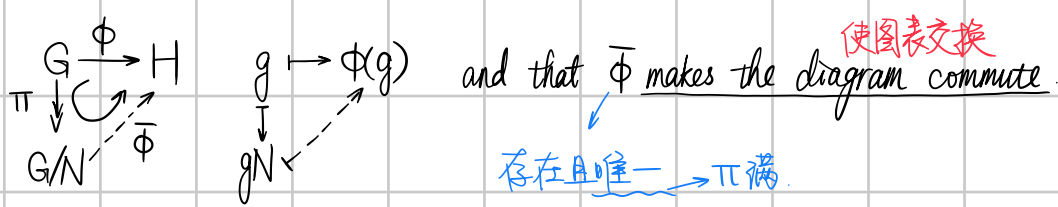
若群同态  $\phi: G \rightarrow H$  满足  $N \subset \ker \phi$ ,

则存在唯一群同态  $\bar{\phi}: G/N \rightarrow H$

满足  $\phi = \bar{\phi} \circ \pi$ .

这说明要考察从  $G$  映出的核包含  $N$  的同态时,

只需考察  $G/N$  即可.



eg. a homomorphism  $\phi: \mathbb{Z} \rightarrow \mathbb{C}^\times$  is determined by  $\phi(1) \in \mathbb{C}^\times$

to get a homomorphism  $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{C}^\times$  need  $n\mathbb{Z} \subset \ker \phi$   $\phi(1)^n$

$$\Leftrightarrow \phi(n\mathbb{Z}) = 1 \Leftrightarrow \phi(n) = 1 \Leftrightarrow \phi(1) \text{ is an } n^{\text{th}} \text{ root of unity.}$$

Recall. fundamental question in group theory, ultimate goal:

classify all groups in isomorphisms.

idea. if  $N \triangleleft G$  can think of  $G$  as built from  $N$  &  $G/N$   $G \cong N + G/N$ .

DEF. A group  $G$  is called simple if  $G \neq \{e\}$  and the only normal subgroups of  $G$  are  $\{e\}$  and  $G$ . 美比素数

eg.  $\mathbb{Z}/p\mathbb{Z}$  for  $p$  a prime number (these are all abelian simple groups).

$A_n$  for  $n \geq 5$  (will see)

$\wedge$   
 $S_n$

REM.  $\exists$  infinite simple groups. eg.  $A_\infty = \bigcup_n A_n$

$\exists$  finitely generated, even finitely presented, infinite simple groups.  $\nearrow$  hard!

Hölder's program (for finite groups)

1) classify all finite simple groups

2) find all ways of "putting simple groups together" to form larger groups.

BIG THM (Classification of finite simple groups)

Every finite simple group is isomorphic to one in

•  $\mathbb{Z}/p\mathbb{Z}$   $p$  prime

•  $A_n$   $n \geq 5$

• 16 families of simple group of Lie type

几类之间有重复 如:  $\text{PSL}_2(\mathbb{F}_9) \cong A_6$ .

eg.  $\text{PSL}_n(\mathbb{F})$   $n \geq 2$   $\mathbb{F}$  finite field

projective special linear group

(except  $\text{PSL}_2(\mathbb{F}_2) \cong S_3$      $\text{PSL}_2(\mathbb{F}_3) \cong A_4$ )

- 26 sporadic simple groups

eg. Monster group

THM (Feit-Thompson). If  $G$  is a finite simple group of odd order, then  $G \cong \mathbb{Z}/p\mathbb{Z}$ .

### Composition series

DEF In a group  $G$ , a sequence of subgroups

$\{e\} = N_0 < N_1 < \dots < N_k = G$  is called a composition series, 合成序列

if  $N_{i-1} \triangleleft N_i$  and  $N_i/N_{i-1}$  is a simple group.

In this case, the  $N_i/N_{i-1}$  are called composition factors

or Jordan-Hölder factors of  $G$ .

REM Do NOT require each  $N_i \triangleleft G$ .

eg.  $\{1\} \triangleleft \langle r^2 \rangle \triangleleft \langle r \rangle \triangleleft D_8$

$\{1\} \triangleleft \langle s \rangle \triangleleft \langle s, r^2 \rangle \triangleleft D_8$

( $\langle s \rangle$  is not normal in  $D_8$ )

all composition factors are  $\mathbb{Z}/2\mathbb{Z}$ .

THM (Jordan-Hölder). Let  $G$  be a finite group  $G \neq \{e\}$ . Then

1)  $G$  has a composition series.

可类比算术基本定理

2) The composition factors are unique.

但群的唯一分解不够强,

ie. if we have two composition series

差一个“作用方式”

$\{e\} = N_0 \triangleleft N_1 \triangleleft \dots \triangleleft N_m = G$

只给定  $N_{i+1}/N_i$  时一般不能唯一确定  $G$ ,

$\{e\} = M_0 \triangleleft M_1 \triangleleft \dots \triangleleft M_n = G$

而正整数却可由其素因子唯一确定

then  $m=n$  and  $\exists$  permutation  $\sigma: \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ , st.  $N_i/N_{i-1} \cong M_{\sigma(i)}/M_{\sigma(i)-1}$ .

Pf. 1) is easy. if  $G$  is simple  $\{e\} \triangleleft G$ . ✓

if not, the  $G$  has a nontrivial normal subgroup  $N$ .

→ reduce to  $N$  and  $G/N$  (by induction on  $\#G$ ).

suppose  $\{e\} = A_0 \triangleleft A_1 \triangleleft \dots \triangleleft A_r = N$

$\{e\} = B_0 \triangleleft B_1 \triangleleft \dots \triangleleft B_s = G/N$  4<sup>th</sup> isom. thm.

⇒  $\{e\} = A_0 \triangleleft A_1 \triangleleft \dots \triangleleft A_r = N = \pi^{-1}(B_0) \triangleleft \pi^{-1}(B_1) \triangleleft \dots \triangleleft \pi^{-1}(B_s) = G$ .

quotient  $\simeq B_i/B_0$  by 3<sup>rd</sup> isom. thm.

2) Toy model:

|                 |                     |
|-----------------|---------------------|
| Set theory      | Group theory        |
| $A$             | $G$                 |
| $A \subset B$   | $H \triangleleft G$ |
| $B \setminus A$ | $G/H$               |

Set-theoretical version.

Let  $X$  be a set with two chains of inclusions

$\emptyset = A_0 \subset A_1 \subset \dots \subset A_m = X$

$\emptyset = B_0 \subset B_1 \subset \dots \subset B_n = X$

Then  $\forall i, j, (A_{i-1} \cup (A_i \cap B_j)) \setminus (A_{i-1} \cup (A_i \cap B_{j-1}))$

$= (B_{j-1} \cup (A_i \cap B_j)) \setminus (B_{j-1} \cup (A_i \cap B_j))$  可画图理解.

Group-theoretical version.

Let  $G$  be a group (not necessarily finite)

suppose given two chains:

$\{e\} = A_0 \triangleleft A_1 \triangleleft \dots \triangleleft A_m = G$

$\{e\} = B_0 \triangleleft B_1 \triangleleft \dots \triangleleft B_n = G$

Then  $\forall i, j, A_{i-1}(A_i \cap B_{j-1}) \triangleleft A_{i-1}(A_i \cap B_j)$

$B_{j-1}(A_{i-1} \cap B_j) \triangleleft B_{j-1}(A_i \cap B_j)$

Moreover  $\frac{A_{i-1}(A_i \cap B_j)}{A_{i-1}(A_i \cap B_{j-1})} \sim \frac{B_{j-1}(A_i \cap B_j)}{B_{j-1}(A_{i-1} \cap B_j)}$

这证明看似非常恶心,但事实上普适性很强

只要所考虑的对象上有三种操作形如交、并、补

且满足此断言,

此处分别为交、积、商

Claim  $\Rightarrow$  Jordan-Hölder 2):

(for non-necessarily finite  $G$ )

Given 2 composition series  $\rightarrow$  get refinements <sup>加细</sup>

$\{e\} \triangleleft A_1 \cap B_1 \triangleleft A_1 \cap B_2 \triangleleft \dots \triangleleft A_1 \cap B_n = A_1 \triangleleft A_1(A_2 \cap B_1) \triangleleft A_1(A_2 \cap B_2) \triangleleft \dots \triangleleft A_1(A_2 \cap B_n) = A_2 \triangleleft \dots \triangleleft G$

exactly one is nontrivial since  $A_1$  is simple say  $A_1 \cap B_{\sigma(1)}$  exactly one is nontrivial since  $A_2/A_1$  is simple say  $A_1(A_2 \cap B_{\sigma(2)})$

Similarly,

$\{e\} \triangleleft A_1 \cap B_1 \triangleleft A_2 \cap B_1 \triangleleft \dots \triangleleft A_m \cap B_1 = B_1 \triangleleft B_1(A_1 \cap B_2) \triangleleft B_1(A_2 \cap B_2) \triangleleft \dots \triangleleft B_1(A_m \cap B_2) = B_2 \triangleleft \dots \triangleleft G$

... say  $A_{\tau(1)} \cap B_1$  say  $B_1(A_{\tau(2)} \cap B_2)$

$\sigma: \{1, \dots, m\} \rightarrow \{1, \dots, n\} \quad \tau: \{1, \dots, n\} \rightarrow \{1, \dots, m\}$

$\frac{A_i}{A_{i-1}} \sim \frac{A_{i-1}(A_i \cap B_{\sigma(i)})}{A_{i-1}(A_i \cap B_{\sigma(i-1)})} \stackrel{\text{Claim}}{\sim} \frac{B_{\sigma(i)-1}(A_i \cap B_{\sigma(i)})}{B_{\sigma(i)-1}(A_{i-1} \cap B_{\sigma(i)})} \sim \frac{B_{\sigma(i)}}{B_{\sigma(i)-1}}$

Moreover,  $\sigma, \tau$  inverse to each other. in particular,  $m=n$ .  $\square$

Pf of Claim.

First, inside  $A_i$ ,  $A_i \cap B_{j-1} \triangleleft A_i$ ,  $A_{i-1} \triangleleft A_i$

$\Rightarrow A_{i-1}(A_i \cap B_{j-1}) \triangleleft A_i$

Moreover  $B_{j-1} \triangleleft B_j \Rightarrow A_i \cap B_{j-1} \triangleleft A_i \cap B_j$

Now check that  $A_{i-1}(A_i \cap B_{j-1}) \triangleleft A_{i-1}(A_i \cap B_j)$

$\begin{matrix} \uparrow & \uparrow \\ \alpha & \beta \end{matrix}$

$A_{i-1} \quad A_{i-1} \triangleleft A_i \quad A_i \cap B_{j-1} \triangleleft A_i \cap B_j$

$\alpha\beta \cdot ab \cdot (\alpha\beta)^{-1} = \alpha\beta ab \beta^{-1} \alpha^{-1} = \alpha \beta \alpha \beta^{-1} \beta b \beta^{-1} \alpha^{-1} \rightarrow A_{i-1} \in A_{i-1}(A_i \cap B_{j-1})$

KEY. Construct

$\frac{A_i \cap B_j}{(A_{i-1} \cap B_j)(A_i \cap B_{j-1})} \sim \frac{A_{i-1}(A_i \cap B_j)}{A_{i-1}(A_i \cap B_{j-1})}$

$\frac{A_i \cap B_j}{(A_{i-1} \cap B_j)(A_i \cap B_{j-1})} \sim \frac{B_{j-1}(A_i \cap B_j)}{B_{j-1}(A_{i-1} \cap B_j)}$

Similarly,  $(A_{i-1} \cap B_j)(A_i \cap B_{j-1})$

$\Phi: A_i \cap B_j \rightarrow A_{i-1}(A_i \cap B_j) \rightarrow A_{i-1}(A_i \cap B_{j-1})$

$\beta \mapsto \beta \xrightarrow{\text{任取一陪集}} \alpha\beta A_{i-1}(A_i \cap B_{j-1}) \xrightarrow{A_{i-1} \quad A_i \cap B_j} \beta \alpha^{-1} A_{i-1}(A_i \cap B_{j-1})$

这二者在集合意义下“交换”(因  $A_{i-1} \triangleleft A_i$ ), 代价是可能需要换个元素.

$\phi$  is surjective.  $\ker \phi = A_i \cap B_j \cap (A_{i-1} (A_i \cap B_{j-1})) \stackrel{\text{obvious}}{=} (A_{i-1} \cap B_j) (A_i \cap B_{j-1})$ .

$$\begin{array}{c} \begin{array}{cc} & ab \\ \swarrow & \nearrow \\ A_{i-1} & A_i \cap B_{j-1} \end{array} \\ \left\{ \begin{array}{l} ab \in B_j \\ b \in B_j \end{array} \right\} \Rightarrow a \in B_j \Rightarrow ab \in (A_{i-1} \cap B_j) (A_i \cap B_{j-1}) \Rightarrow "C" \checkmark. \end{array}$$

$\Rightarrow \ker \phi = (A_{i-1} \cap B_j) (A_i \cap B_{j-1})$ . By the 1<sup>st</sup> isom. thm, ...  $\square$  任意列群的有限性只用在合成序列的存在性证明中.

